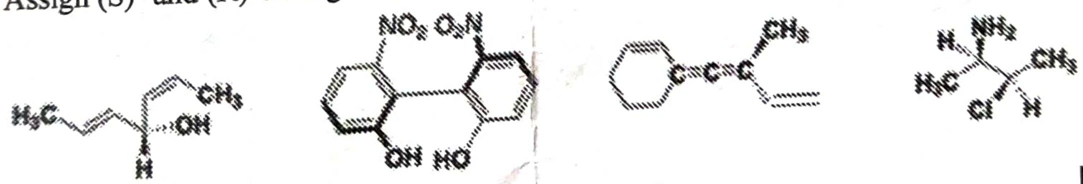
Mid-Semester Examination

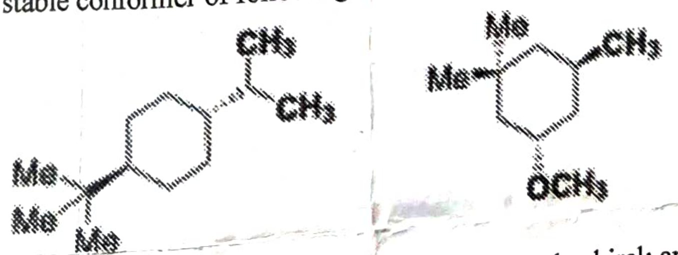
Introduction to Fundamental and Applied Chemistry-I

Date: 20 September 2025 Total Marks: 30 Time: 90 minutes

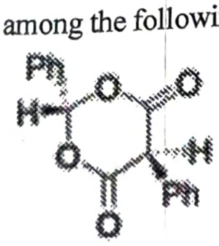
1. Find out the dimension of a *3*-dimensional wavefunction. [2]
2. What is the uncertainty in velocity, if we wish to locate an electron within an atom, say, so that ∇x is approximately 50 pm? [2]
3. Write the kinetic-energy operator $\hat{T}$ for a two-particle, 3-dimensional system. [2]
4. Arrange the levels and states of ^3P term that arises from a p^2 configuration. [2]
5. Calculate the degeneracy of the following terms: [2]
   (a) ^{3}F (b) ^{1}G (c) ^{2}D (d) ^{4}P
6. A particle undergoes oscillatory motion in the range from -x to +x around its equilibrium position, x=0. Find the average values for the position and linear momentum of the particle. The wavefunction for the system is $\varphi_0(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$. [[illegible]]
7. Find the total angular momentum of an electron, when it is in: [3]
   (a) s-orbital (b) p-orbital (c) d-orbital
8. Write the ground state term symbol (in the presence of L-S coupling) for Si, P, S and Cl atoms. [4]
9. When lithium is irradiated with light, the kinetic energy of the ejected electrons is 2.935×10^{-19} J for $\lambda = 300$ nm and 1.28×10^{-19} J for $\lambda = 400$ nm. Calculate the Planck constant, the threshold frequency, and the work function of lithium from these data. [4]
10. Assign (S)- and (R)-configurations to the following molecules. [2]



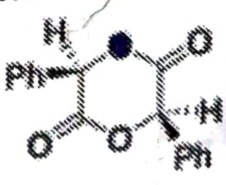
11. Draw the most stable conformer of following two molecules. [2]



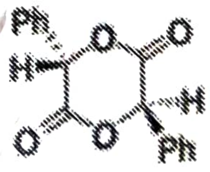
12. Identify among the following molecules which one is chiral and achiral; and why? [3]



A



B



C